SPA617 — RF Components: Characterization and Measurement

Experiment: Splitter Characterization

1. Objective

To characterize a two-way RF power splitter by measuring the output powers at different frequencies and determining the insertion/split loss of the two output ports.

2. Components Required

Two-way RF power splitter, RF signal source, power meter/spectrum analyzer or VNA, RF cables and connectors, and suitable 50 Ω terminations.

3. Theory

A power splitter is a passive RF device that divides one input signal into two or more output signals. For an ideal two-way equal power splitter, the input power is divided equally between the two output ports.

For an ideal two-way splitter, the voltage division factor is approximately $1/\sqrt{2} = 0.707$ and the ideal power split corresponds to approximately 3 dB loss at each output.

The measured loss at each output port is calculated from:

$$\mathbf{Loss_1\ (dB) = P_{in} - P_{out1}}$$

$$\mathbf{Loss_2\ (dB) = P_{in} - P_{out2}}$$

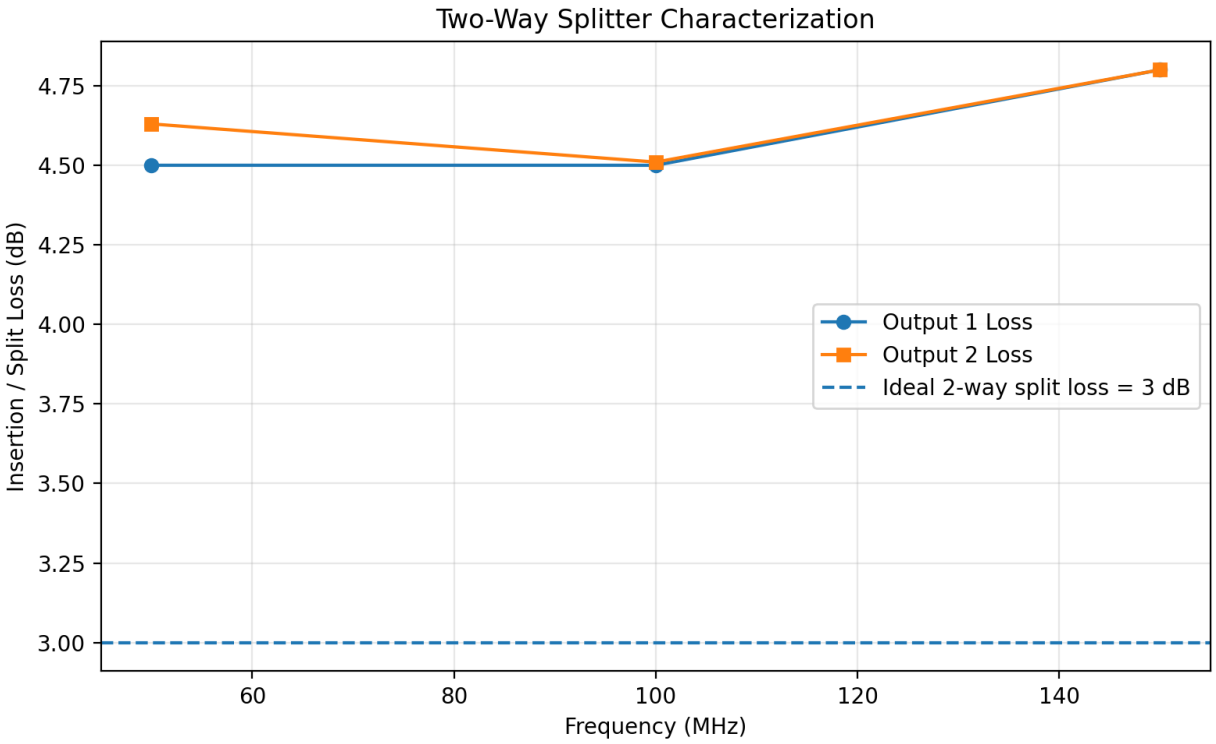
where P_{in} is the input power, while P_{out1} and P_{out2} are the powers measured at output ports 1 and 2, respectively.

4. Experimental Setup

The RF source was connected to the input port of the two-way splitter. Output ports 1 and 2 were connected to the measuring instrument through suitable RF cables. The input power was maintained at 0 dBm and measurements were taken at 50 MHz, 100 MHz and 150 MHz.

5. Observation & Graphs

Frequency (MHz)	P_in (dBm)	P_out1 (dBm)	P_out2 (dBm)	Loss1 (dB)	Loss2 (dB)
50	0.00	-4.50	-4.63	4.50	4.63
100	0.00	-4.50	-4.51	4.50	4.51
150	0.00	-4.80	-4.80	4.80	4.80


Two-Way Splitter Characterization
Insertion / Split Loss (dB)
4.75
4.50
4.25
4.00
3.75
3.50
3.25
3.00
60
80
100
120
140
Frequency (MHz)
Output 1 Loss
Output 2 Loss
Ideal 2-way split loss = 3 dB

6. Calculation & Results

The input power was maintained at P_{in} = 0 dBm.

At 50 MHz: $Loss_1$ = 0 – (–4.50) = **4.50 dB**; $Loss_2$ = 0 – (–4.63) = **4.63 dB**.

At 100 MHz: $Loss_1$ = 0 – (–4.50) = **4.50 dB**; $Loss_2$ = 0 – (–4.51) = **4.51 dB**.

At 150 MHz: $Loss_1$ = 0 – (–4.80) = **4.80 dB**; $Loss_2$ = 0 – (–4.80) = **4.80 dB**.

The two output ports show nearly similar losses, indicating approximately equal power division. The measured loss is around 4.5–4.8 dB, which is higher than the ideal 3 dB split loss; the additional loss can be attributed to practical device and measurement/cable losses.

7. Result

The two-way RF splitter was characterized successfully. Both output ports receive approximately equal signal power. The measured split loss is approximately 4.5–4.8 dB over the measured 50–150 MHz range.

8. Conclusion / Key Takeaway

The experiment demonstrates the operation of a two-way RF power splitter. An input RF signal is divided between two output ports with nearly equal output levels. The measured output losses are close to each other, confirming balanced power division. The measured loss is greater than the ideal 3 dB because a practical splitter has additional insertion, connector and cable losses.

Note: The observation values used in this report are transcribed from the provided laboratory observation sheet.